GERMANY'S ROLE IN GLOBAL CLIMATE MITIGATION:

EMISSIONS, RESPONSIBILITY, AND CARBON INEQUALITY

(1990-2024)

Anjali Arya
(m2024bsass004)
m2024bsass004@stud.tiss.ac.in

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Introduction

Germany, officially known as the Federal Republic of Germany, holds a unique status in worldwide climate discussions. The country controls substantial economic and political power because it serves as Central Europe's biggest market and European Union's most populous country with 84.6 million residents in 2024 and as the third most powerful economy based on nominal GDP[1]. The country has earned international recognition for its high-tech manufacturing sector which includes automotive engineering and chemical production and precision machinery and for its growing influence in international energy and environmental decision-making.

Germany maintains political stability through its federal parliamentary democracy system which establishes its status as a founding member of both the European Union and the Eurozone. The Human Development Index (HDI) score of 0.95 positions the country among the ten most developed nations worldwide which increases both its ability and duty to address climate change issues.

Germany serves as an interesting study because its modern historical development shows unresolved conflicts which shaped its entire historical path. The industrial strength of Germany during the twentieth century depended on its extensive supplies of hard coal and lignite (brown coal), which turned the country into one of the most significant greenhouse gas producers among developed nations. Germany started its major energy system transformation through the *Energiewende policy framework* which began in the 1990s and achieved its highest development stage during the early 2000s. This framework commits Germany to greenhouse gas emission reductions which will reach 85 to 90 percent under 1990 emission levels by 2050 and the country will achieve complete climate neutrality by 2045 while it will shut down all coal power plants by 2038 and generate 80 percent of its electricity from renewable energy sources.[2]

The *Energiewende* serves as an energy policy which transforms the entire energy system of a high-income export economy by establishing methods to achieve economic expansion without increasing carbon emissions. The project aims to develop an energy system which relies on renewable sources for its energy supply while solar power and wind power technologies provide electricity through advanced methods of energy efficiency and new systems of electrical generation and distribution. The report studies Germany's CO2 and greenhouse gas emissions from 1990 until 2024 to show the results of progress and remaining challenges and their effects on climate justice in a world which faces rapid temperature increases.[3]

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[1] (*Countryeconomy, 2019)*

[2] *(Germany's CO_2 Emissions Drop to Record Low but Reveal Gaps in Country's Climate Policies, 2024)*

[3] (*Energiewende*, 2020)

Data and Source

This study covers Germany from 1990 to 2024, capturing the pre-Energiewende baseline through the full post-reunification emissions trajectory. For comparative analysis, data were also gathered for the **United States** (High per-capita emitter and largest developed economy), **China** (World's largest total emitter and rapidly industrialising economy), **France** (European neighbour with nuclear-dominated energy mix), **India** (Large developing economy with low per-capita emissions), the **overall Europe**, and **World data**.

All data are taken from *Our World in Data (OWID)* and the *Global Carbon Project (GCP).*

Variables Selected

- **Core variables**: Country, year, ISO code, population.
- **Economic indicators**: GDP, GDP per capita, CO_2 emissions per GDP (kg per international-$).
- **Emissions metrics**: Annual CO_2 emissions, CO_2 per capita, total GHG excluding land use (fossil fuels and industry), total GHG including land use.
- **Disaggregated emissions**: CO_2 by fuel source (coal, oil, gas, cement).
- **Energy data**: Primary energy consumption.
- **Global significance**: Share of global annual CO_2 emissions, share of contribution to global warming, consumption-based CO_2 emissions.

All data cleansing, transformation, and statistical analyses were done in R and PowerBI. Interactive visualisations and trend dashboards were developed in Power BI.

Limitation: GDP data for Europe, World and all the individual countries was not available for 2023 and 2024, so GDP-based indicators (GDP per capita, CO_2 per GDP) are analyzed only for 1990–2022. Emission based analysis (annual CO_2, CO_2 per capita, GHG totals, share of global emissions) is available for the time frame 1990-2024.

Emissions Trend Analysis

Germany demonstrates exceptional environmental achievement because its emission reductions have achieved permanent records. The country achieved a 50% decrease in its per capita emissions which decreased from 13.23 tonnes in 1990 to 6.77 tonnes in 2024, which no other major economy has matched. The US has decreased its emissions at a slower pace, while China has increased its per capita emissions.

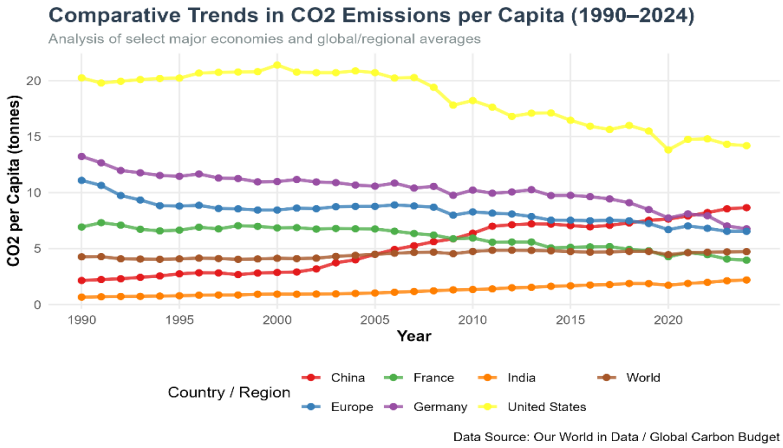

Figure 1

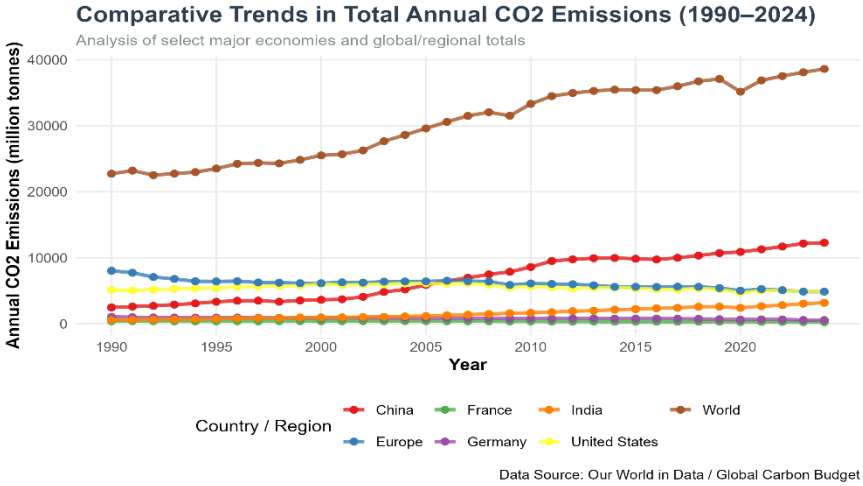

Figure 2

Germany decreased its Total emissions from approximately 1,050 Mt to approximately 560 Mt, whereas China became the world's top emitter and India followed a similar trend. The situation demonstrates a fundamental worldwide conflict because developed nations are cutting their emissions, but emerging economies continue to expand.

Germany achieved a avg GDP per capita of 37183.76 between 1990 and 2022 while the country produced avg per capita emissions of 10.25 tonnes, which fell short of the United States emissions rate. France achieved lower emissions, while maintaining the same

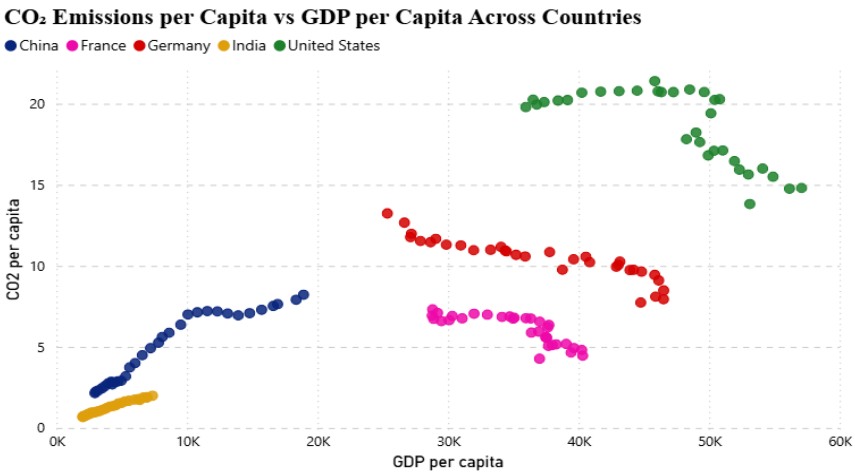

Figure 3

economic development level as Germany which shows that the country needs to achieve more decarbonisation from its Energiewende program.

Germany's Emissions Decline: Key Drivers Behind the Transition

1. **Post-Reunification "Wall Fall" Effect (1990s)**
   The complete shutdown of unproductive businesses in East Germany after German reunification resulted in automatic CO_2 emissions decreases which scientists describe as "wall fall profits" according to Clean Energy Wire. The 1990 baseline for Germany became inflated through this process which made percentage reductions appear greater than their actual impact. The genuine structural transformation occurred at a later time.[4]
2. **Renewables revolution**
   Since Germany's Renewable Energy Act (EEG) launched in 2000, renewables grew from a niche to dominate the electricity mix, covering 56% of power consumption by 2025.
3. **Energy transformation and coal collapse**
   Germany has reduced its energy sector emissions by approximately 60% since the year 1990. The initial emissions reductions happened because Germany shut down its most polluting lignite power plants and installed newer, more efficient power generation technologies. The energy market has witnessed a recent trend where renewable sources of energy have taken over coal as the primary energy source.
   The shift between these two periods turned into an urgent situation which required immediate action during the years 2023 to 2024. Germany reached its lowest emissions level in 2023 when the total emissions reached 673 Mt which marked the lowest point in the last 70 years. The year 2024 continued the decline because coal reductions prevented 20 Mt of emissions during the initial three quarters of the year.[5]
4. **Economic shocks**
   Germany experienced significant reductions in emissions during its major crises when emissions fell by 6.9% in 2009 and 9% in 2020 and about 10% during the 2023 energy crisis. The emissions decline results from decreased industrial operations which do not represent sustainable clean energy progress. This means emissions could rise again if economic activity recovers.

Key Challenges in the Transition

The complete emissions reduction process has achieved major success but certain sectors continue to present challenges. Transport emissions have shown minimal progress since the 1990s while 2024 marks the highest total of 49 million registered cars. The building industry has experienced complete operational standstill since 2011, which jeopardizes their climate-neutral target achievement. The nuclear phase-out process of Germany after the Fukushima

[4] *(Appunn et al., 2025)*

[5] *(German Coal Use Continues Downward Trend in 2024, 2024)*

disaster continues to generate debate because it decreased greenhouse gas emissions through increased coal consumption.[6]

Carbon Inequality Analysis

Average Emissions Profile (1990–2024)

Comparison of Per Capita vs. Total Annual CO2 Emissions

Country / Region	Avg. CO2 per Capita (tonnes)	Avg. Annual CO2 (million tonnes)
World	4.46	30,421.86
China	5.15	6,976.44
Europe	8.29	6,102.05
United States	18.42	5,499.87
India	1.28	1,589.89
Germany	10.25	840.13
France	5.98	368.08

Figure 4

The emission patterns of Germany demonstrate both a decreasing trend and a change in worldwide significance. Germany's per capita emissions from 1990 to 2024 showed an average of 10.25 tonnes which decreased over the years because of energy transition and efficiency improvements and policy implementation yet experienced temporary declines because of economic disruptions.

And the global emission share decreased from approximately 4.7 percent to about 1.5 percent while the country maintained an average of 2.92 percent during that timeframe. The domestic emission reductions exist together with the fact that China and India experienced fast emission increases which caused worldwide emissions to rise and diminished Germany's power to cut emissions.

Greenhouse Gas Emissions Analysis

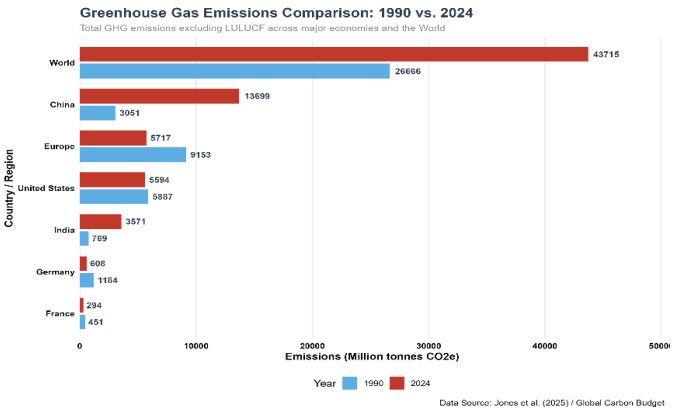

Figure 5

Germany's total greenhouse gas (GHG) tells a more complete story. According to the German Federal Environment Agency (UBA) and the Expert Council on Climate Issues, Germany's total GHG emissions in 2024 fell to 649 million tonnes of CO_2 equivalents (Mt CO_2-eq) , a reduction of 23 million tonnes (3.4%) compared to 2023.

[6] *(Gabbatiss, 2025)*

The GHG data reveals that Germany is on track to meet its 2030 target of 65% reduction below 1990 levels "by a narrow margin" according to the Expert Council's assessment, but this depends on full implementation of existing measures. However, the Expert Council warns that current reduction rates are insufficient, and emissions must decline significantly faster from 2024 onward to meet legal targets.

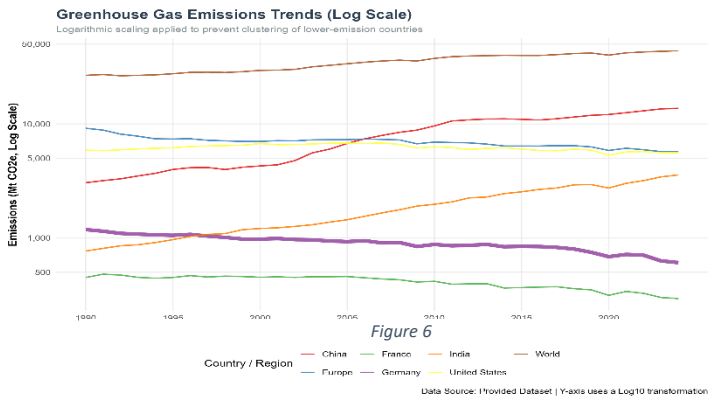

Figure 6

Share of Global Emissions

Germany's global CO_2 emissions share decreased from approximately 6% in 1990 to about 1.48% in 2024 which resulted in an average share of 2.92% between 1990 and 2024. The information presents a minor impact but it actually contains two important aspects which need consideration.

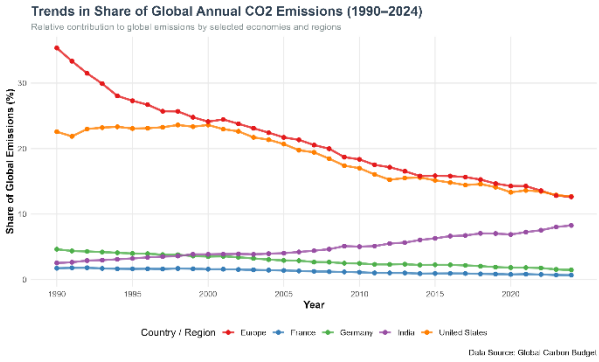

Figure 7

First, Germany's historical responsibility is much larger. The country developed into an early industrialized nation which resulted in significant greenhouse gas emissions. Developed nations assume increased climate responsibilities according to the Common but Differentiated Responsibilities (CBDR) principle. Germany's share of 1.5% current emissions presents an inaccurate assessment of its climate obligations.

Germany holds its position as the EU's top coal producer and consumer while coal generates approximately 21% of electricity in 2025. This situation causes Germany to maintain higher emissions intensity compared to France which shares similar economic output.

Comparison with India and Global Averages

Germany emits 10.25 tons of CO_2 for each citizen which amounts to eight times India's emission of 1.28 tons while Germany's emissions exceed worldwide standards by 2.3 times. India's national emissions currently exceed Germany's emissions because India has 1.4 billion citizens while Germany has 84 million residents. Between 1990 and 2020, Germany's GHG emissions remained flat at approximately 1,100 Mt while India's emissions increased from 800 Mt to 1,500 Mt. From 1990 to 2024, Germany cut 434 Mt (39.5%), but India added 3,572 Mt (446%), Germany's reductions are overwhelmed by India's growth 8:1.

Germany's decarbonization process moves at a slow pace which causes climate injustice because the nation begins from its historical high emissions while India must increase its emissions for necessary progress. Germany produces more emissions than India on a per capita basis yet India exceeds Germany in total atmospheric emissions which grow at a faster rate.

Germany needs to support India's transition to renewable energy sources because of climate justice principles.[7]

High, Moderate, or Low Emitter?

On a per capita basis, Germany is a **high emitter** country even at its 2023 low (8.1t per person), it is nearly double the global average, while countries like France (~5.98t) show lower-emission pathways through nuclear energy.

In total emissions terms, Germany is a **moderate emitter**. With about 2.92% of global emissions and a population of 84 million, it exceeds its population share but remains far below major emitters like China (~32%), the US (~12%), and India (~8%).

In terms of trend, Germany is an **exceptional decliner**, because the country has reduced emissions by approximately 46% since 1990 which resulted in emissions reaching a 70-year low of 673 Mt during 2023.

Climate Responsibility Implications

Germany's position as a high per-capita, historically large emitter with strong economic capacity creates a clear set of obligations under international climate frameworks:

Domestic obligations: Germany's climate targets and policies and its financial systems receive an Insufficient rating because they fail to meet the Paris Agreement 1.5°C target and need major enhancements. The transport sector experienced its worst performance when car ownership reached 49 million in 2024 and buildings continued to rely on fossil gas and oil for heating needs.[8]

financial and technological responsibility: Germany needs to give climate finance and technological support because its historical emissions record combined with its economic strength shows it should take these responsibilities. The country showed its ability to assist cleaner energy adoption in India through its renewable energy growth which increased from less than 2 percent in 2000 to approximately 54 percent in 2024.[9]

The equity paradox: India's per capita emissions (~1.28t) are about one-eighth of Germany's (~10.25t), yet India faces severe climate impacts. This highlights a core justice issue: countries with higher historical and per capita emissions must bear a greater share of climate action.

Climate Justice Reflection

The emissions data from Germany shows that the country has decreased its total CO_2 emissions by 46% since 1990 yet still maintains a high per-capita emission rate of 10.25 tonnes which is eight times higher than India's 1.28 tonnes. Report correctly identifies Germany's historical

[7] (*The Global Contributors to Climate Change • Environmental Studies (EVS) Institute*, 2025)

[8] *(Germany | Climate Action Tracker, 2024)*

[9] *(International Climate Finance 2024: Germany Remains Reliable Partner in Global Climate Policy, 2024)*

responsibility but the International Court of Justice's advisory opinion from July 2025 creates a new requirement for Germany.

The ICJ has established that the Paris Agreement's 1.5°C target functions as a legally binding obligation under international human rights law. For Germany, this creates a staggering reality: with just 1% of the world's population, it has already exhausted its per-capita carbon budget. The Court determined that historical polluters need to pay vulnerable nations according to their proportion of past emissions which equates to about 5% for Germany.

Germany's role, therefore, is no longer voluntary. It must:

- First, The 5% liability needs to be transformed into Loss and Damage contributions which will become legally binding through this process. Germany committed to provide €6 billion in climate finance each year until 2025 but actual funding will decrease to €4.5-5.3 billion during the 2025-26 period which constitutes a breach of credibility.[10]
- Second, Germany is currently constructing a green hydrogen hub which will cost €1.3 billion in Andhra Pradesh, India. The funding system requires co-investment because it needs to be expanded beyond its current form.[11]
- Third, the new German government plans to construct 20 gigawatts of new gas power facilities while it cancels electric vehicle targets which results in an "Insufficient" climate performance rating from Climate Action Tracker.

Mitigation and adaptation are complements, not substitutes. PANTA RHEI modelling shows that without strong mitigation, adaptation measures can become ineffective. Countries like India emitting just 1.28 tonnes per capita compared to Germany's 10.25 but they are on the frontline of adaptation despite contributing little to cumulative emissions. Every tonne Germany fails to reduce increases the adaptation burden on such nations. Germany's role is therefore clear: complete its own transition, finance the Global South, and export the technologies needed for global decarbonisation.

Conclusion

The climate path taken by Germany since its reunification until today demonstrates both successful achievements and incomplete efforts. The Energiewende shows that an industrial nation can achieve complete decarbonization without losing its industrial base. The electricity sector achieved success yet this success created an illusion of progress while transportation and building operations remained stagnant.

The ICJ's July 2025 ruling changes the stakes entirely. Germany's historical emissions have become legal responsibility instead of moral issue. The era of voluntary pledges is ending; the era of binding obligations has begun. The data proves three things. First, mitigation is

[10] (*Germany Spent €6.1bln on International Climate Finance Last Year, but 2025 Target "Great Challenge,"* 2025)

[11] *(Germany Pledges €1.3 Billion to India for Climate, Energy and Sustainable Development Cooperation | NetZero India, 2025)*

economically rational, PANTA RHEI modelling shows positive GDP effects from climate action. Second, Germany's transport and building sectors prove that policy architecture without enforcement delivers nothing. Third, adaptation without mitigation is futile. Each tonne Germany fails to reduce creates costs for vulnerable nations such as India.

Germany faces two options. Either the country can maintain its leadership claims but continue building new gas plants while reducing its climate funding or the country must acknowledge its 5% historical climate responsibility because it needs to pay its climate debt and share essential technologies with the world.

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